

Your grade: 96%

Your latest: 96% • Your highest: 96% • To pass you need at least 80%. We keep your highest score.

Next item →

1. Which of the following statements can be expressed using predicate logic with appropriate quantifiers?

0.8 / 1 point

- ☐ This sentence is false.
- ☐ The sun will rise in the west tomorrow.
- ☒ Some birds can fly.

✔ Nice work

Well done! This can be expressed using an existential quantifier. Reflect on how quantifiers help in forming such assertions.

- ☒ All cats are mammals.

✔ Nice work

Correct! Predicate logic can express this statement using a universal quantifier. Consider how predicates work with quantifiers in this context.

- ☐ Only one person knows the solution to this problem.

⊗ You didn't select all the correct answers

2. In predicate logic, how does the existential quantifier differ from the universal quantifier?

1 / 1 point

- ☒ Existential quantifier asserts that a property holds for at least one element in the domain.
- ☐ Existential quantifier asserts that a property holds for all elements in the domain.
- ☐ Existential quantifier asserts a property holds for a majority of elements in the domain.
- ☐ Existential quantifier ensures a property holds for exactly one element in the domain.

✔ Nice work

Great job! The existential quantifier indeed confirms that there is at least one element for which the property holds. Review examples to solidify your understanding.

3. Which of the following statements best distinguishes between propositional logic and predicate logic?

1 / 1 point

- ☐ Predicate logic only deals with concrete facts, while propositional logic deals with abstract concepts.
- ☐ Propositional logic requires more complex syntax than predicate logic.
- ☒ Propositional logic deals with propositions as units, while predicate logic deals with predicates and quantifiers.
- ☐ Propositional logic uses quantifiers extensively, while predicate logic does not.

✔ Nice work

Correct! Propositional logic focuses on whole propositions, whereas predicate logic involves predicates and quantifiers for more complex statements.

4. Which of the following are valid applications of rules of inference to the statement: "If all humans are mortal and Socrates is a human, then Socrates is mortal"? Select all that apply.

1 / 1 point

- ☐ Hypothetical Syllogism
- ☒ Modus Ponens

✔ Nice work

Correct! Modus Ponens is used when you have a conditional statement and its antecedent, allowing you to conclude the consequent.

- ☒ Universal Instantiation

✔ Nice work

Good job! Universal Instantiation allows you to apply a general principle to a specific case, such as applying 'all humans are mortal' to Socrates.

- ☐ Existential Generalization
- ☐ Modus Tollens

5. Which of the following expressions correctly applies De Morgan's laws to negate the quantified statement:

1 / 1 point

$$\forall x(P(x) \wedge Q(x))$$

?

☐
$$\forall x(\neg P(x) \vee \neg Q(x))$$

☐
$$\forall x(\neg P(x) \wedge \neg Q(x))$$

☐
$$\exists x(P(x) \vee Q(x))$$

☒
$$\exists x(\neg P(x) \vee \neg Q(x))$$

✔ Nice work

Great job! Remember, De Morgan's laws change the quantifiers when negating logical expressions: $\forall$ becomes $\exists$ and $\wedge$ becomes $\vee$.